Climate-Driven Maize Yield Instability Across Global Climate Panels: A Data-Driven Clustering Approach Using CY-Bench

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***Abstract*—Building on a preliminary investigation of European maize yield panels using theory-based Köppen-Geiger climate classifications, this study extends the analysis to a global scale by replacing pre-defined climate zones with data-driven clustering. Specifically, climate-driven maize yield instability is investigated across 42 countries using the CY-Bench global subnational dataset (2001–2023), comprising 157,894 region-year observations. A key methodological contribution is the replacement of theory-based climate zone classifications with data-driven k-means clustering directly on growing-season climate variables. Hemisphere-aware feature engineering ensures that heat stress and precipitation predictors reflect actual growing-season conditions rather than calendar-year artefacts. A Kruskal-Wallis test confirms that yield instability differs significantly across climate clusters ($H = 1195.22$, $p < 0.001$). A tuned Random Forest model outperforms a naive zero-prediction baseline in two of three clusters, with statistical significance confirmed via Wilcoxon signed-rank tests ($p < 0.05$). Partial dependence analysis reveals non-linear threshold effects: yield drops sharply beyond approximately 50 heat stress days per growing season, and suffers most when growing-season precipitation falls below 600 mm. These findings highlight that climate adaptation strategies for maize must be cluster-specific rather than globally uniform.**

***Index Terms*—maize yield instability, climate clustering, Random Forest, CY-Bench, partial dependence plots, hemisphere-aware features**

I. Introduction

Global maize production faces increasing pressure from climate variability, with yield instability threatening food security across both temperate and tropical regions [1]. While the relationship between climate and crop yields has been extensively studied, most existing work either focuses on single regions or relies on pre-defined climate zone classifications that may not align with the agricultural phenomena under investigation [2], [3].

A persistent challenge in global crop-climate studies is the treatment of geographic heterogeneity. Theory-based classifications such as the Köppen-Geiger system group countries by long-run climatological archetypes that are not constructed from the same variables used in yield modelling. This mismatch can obscure cluster-specific climate-yield relationships that are critical for targeted adaptation policy [4].

Machine learning approaches have demonstrated promise in capturing non-linear climate-yield relationships [5], [7], yet hemisphere-specific growing season definitions and global subnational coverage remain underexplored. Studies such as Harsányi et al. [6] and Lecerf et al. [3] provide valuable regional insights but do not extend to a multi-hemisphere global framework.

This study makes four contributions. First, hemisphere-aware growing-season feature engineering is applied to a global subnational dataset, ensuring climate predictors reflect actual crop-growing conditions. Second, theory-based climate panels are replaced with data-driven k-means clustering and the grouping is empirically validated. Third, non-parametric statistical tests (Kruskal-Wallis, Wilcoxon signed-rank) that make no distributional assumptions about yield variability are employed. Fourth, partial dependence plots are used to reveal non-linear, threshold-based climate-yield relationships that linear models would miss.

II. Data and Methods

A. Dataset

The CY-Bench global subnational maize dataset [8] is used, covering 42 countries at the administrative unit level for the period 2001–2023. After quality filtering (minimum 5 observations per region for trend estimation), the working dataset contains 157,894 region-year observations across 38 countries.

B. Hemisphere-Aware Feature Engineering

Two climate features are constructed with hemisphere-specific growing-season windows. For Northern Hemisphere countries (NH), heat stress days are defined as the count of days exceeding 30°C during June–August, and growing-season precipitation accumulates over April–September. For Southern Hemisphere and year-spanning growing seasons, the corresponding windows are December–February and October–March, respectively. This ensures that predictors capture stress during the actual crop growth period rather than calendar-year averages.

C. Yield Detrending

Ordinary least squares (OLS) linear trends are estimated per administrative region (minimum 5 observations). Detrended yield is the residual after removing the fitted trend, representing inter-annual variability net of long-run productivity growth. Yield instability per region is quantified as the coefficient of variation (CV) of detrended yield.

D. Data-Driven Climate Clustering

A preliminary analysis grouped nine European countries into three theory-based climate panels (Mediterranean, Temperate, Continental) following the Köppen-Geiger classification. However, since these zones are defined by long-run climatological archetypes unrelated to the study variables, a data-driven approach was adopted for the global analysis. Prior to clustering, the distribution of both features was examined. Both heat stress days (skewness = 1.05) and growing-season precipitation (skewness = 1.54) are right-skewed and non-normal (Shapiro-Wilk $p < 0.001$), confirming that standardisation prior to k-means is essential to prevent scale dominance. Countries are clustered using k-means ($k = 3$) on standardised country-level mean heat stress days and mean growing-season precipitation. The number of clusters is selected via elbow method and silhouette analysis: $k = 3$ achieves the highest silhouette score (0.497) across all values of k from 2 to 8. Ward hierarchical clustering is applied as a robustness check: 37 of 38 countries receive identical cluster assignments, confirming stability (silhouette scores: k-means = 0.497, hierarchical = 0.496).

E. Statistical Comparison of Yield Instability

In the preliminary analysis, a Kruskal-Wallis test applied to the three Köppen-Geiger panels (Mediterranean, Temperate, Continental) yielded a significant result, confirming that yield instability differs across theory-based panel groupings. The same test is applied here to the data-driven clusters to allow a direct methodological comparison. A Kruskal-Wallis H-test [9] is applied to test whether CV of detrended yield differs across clusters. Pairwise post-hoc comparisons use the Dunn test [11] with Bonferroni correction.

F. Machine Learning Models

In the preliminary analysis, only a naive baseline was established for the three Köppen-Geiger panels using nine European countries. In the final analysis, Decision Tree (DT) and Random Forest (RF) regressors are trained on both the Köppen-Geiger panels and the data-driven clusters, enabling a direct comparison between the two regionalisation approaches. Brazil and the United States are excluded from ML training due to their disproportionate representation within their respective clusters (98% and 81% of cluster observations), leaving 36 countries for training. The preliminary analysis considered four candidate climate predictors: heat stress days, growing-season precipitation, climatic water balance deficit (`cwb_deficit_gs`), and mean growing-season temperature (`tavg_jja`). A variance inflation factor (VIF) analysis revealed severe multicollinearity in two of the four features (`tavg_jja`: VIF = 130.0; `cwb_deficit_gs`: VIF = 80.4), so these were dropped. The remaining two features; heat stress days (VIF = 3.96) and growing-season precipitation (VIF acceptable after removal of the collinear predictors) were retained as the final ML predictors and clustering variables. Models are evaluated using relative root mean squared error (rRMSE = RMSE / mean yield $\times$ 100). A naive baseline predicts detrended yield = 0 for all observations. Hyperparameters are tuned via GridSearchCV with 5-fold cross-validation; the best parameters identified are: *max_depth* = 3, *min_samples_leaf* = 20, and *n_estimators* = 200.

G. Statistical Significance Testing

A one-sided Wilcoxon signed-rank test [10] evaluates whether the DT and RF models' absolute errors are significantly smaller than those of the naive baseline per cluster. The test is applied independently for each model–cluster combination.

H. Interpretability

Global and per-cluster permutation importance identifies which feature drives predictions. Partial dependence plots (PDPs) reveal the marginal effect of each feature on predicted detrended yield in original units (days and mm).

III. Results

A. Climate Clusters

Three climate clusters emerge from k-means (Fig. 1): Cluster 0 (semi-arid, $n = 9$ countries) is characterised by high heat stress and moderate precipitation; Cluster 1 (temperate, $n = 24$ countries) by low heat stress and moderate precipitation; and Cluster 2 (tropical, $n = 5$ countries) by moderate heat stress and high precipitation.

B. Yield Instability Across Clusters

The Kruskal-Wallis test strongly rejects the null hypothesis of equal yield instability across clusters ($H = 1195.22$, $p = 2.90 \times 10^{-260}$). The tropical cluster exhibits the highest mean CV (27.5%), followed by the semi-arid cluster (19.4%) and the temperate cluster (20.3%) (Fig. 2). All pairwise Dunn post-hoc comparisons are statistically significant after Bonferroni correction.

Examining each cluster in turn reveals distinct instability profiles. Cluster 0 (semi-arid, 9 countries, median CV = 17.8%) shows moderate central instability but a wide interquartile range, reflecting the heterogeneous membership: Sahelian countries such as Niger, Mali, and Senegal experience highly erratic rainfall, while Argentina and the United States benefit from irrigation and advanced agronomic management that partially buffers inter-annual variance. Cluster 1 (temperate, 24 countries, median CV = 17.4%) is the most

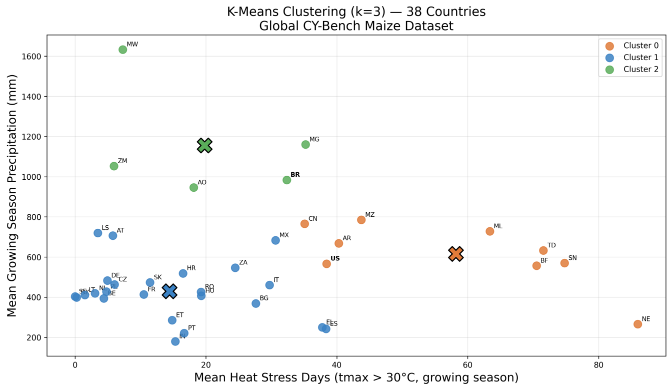

Fig. 1. K-means climate clusters ($k = 3$) in heat stress days vs growing-season precipitation space. Each point represents a country's mean growing-season climate profile.

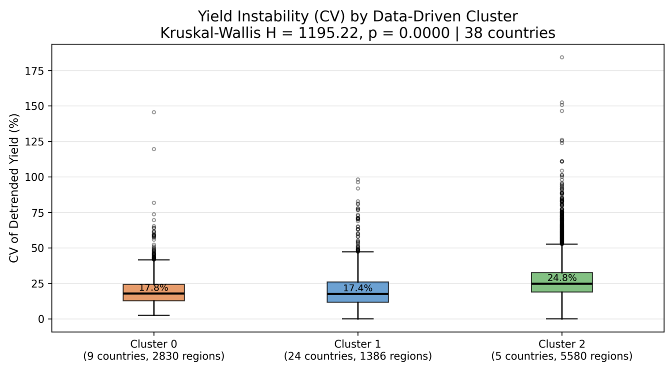

Fig. 2. Coefficient of variation (CV) of detrended maize yield per climate cluster. The tropical cluster (Cluster 2) shows the highest instability. Kruskal-Wallis $H = 1195.22$, $p < 0.001$.

stable cluster with the tightest distribution, consistent with the relatively predictable growing-season climate of European maize-producing regions and the moderate heat and precipitation conditions that define this group. Cluster 2 (tropical, 5 countries, median CV = 24.8%) exhibits both the highest central tendency and the widest spread. Angola, Madagascar, Malawi, Mozambique, and Zambia all face pronounced inter-annual rainfall variability, and the wide box reflects the fact that instability levels differ substantially even within this small group of five countries.

C. *Predictive Performance and Statistical Significance*

Table I summarises model performance. Both the tuned DT and tuned RF outperform the naive baseline in Clusters 0 and 1, with Wilcoxon signed-rank tests confirming statistical significance ($p < 0.05$) for both models. The RF achieves the lowest rRMSE in each significant cluster (Cluster 0: 23.9%; Cluster 1: 18.4%), marginally outperforming the DT (Cluster 0: 24.1%; Cluster 1: 18.4%). In Cluster 2, neither model significantly outperforms the baseline, attributable to the small training set (approximately 1,400 observations across four countries after Brazil's exclusion).

TABLE I
MODEL PERFORMANCE PER CLIMATE CLUSTER. WILCOXON RESULT IS IDENTICAL FOR BOTH DT AND RF (ONE-SIDED SIGNED-RANK TEST VS NAIVE BASELINE, $p < 0.05$).

Cluster	Naive rRMSE	DT rRMSE	RF rRMSE	Wilcoxon (DT & RF)
0 (Semi-Arid)	25.3%	24.1%	23.9%	$p < 0.05$
1 (Temperate)	18.7%	18.4%	18.4%	$p < 0.05$
2 (Tropical)	31.6%	33.1%	33.3%	n.s.

D. *Theory-Based vs Data-Driven Clustering*

Fig. 3 contrasts performance under theory-based Köppen panels (9 European countries, tuned DT and RF) against data-driven clusters (36 countries, tuned DT and RF). Under theory-based panels, the RF improves on the naive baseline in 2 of 3 panels (Temperate: 12.2% vs 13.4%; Continental: 22.9% vs 24.0%), with the Mediterranean panel performing marginally worse than naive (17.4% vs 17.2%). Under data-driven clustering, 2 of 3 clusters also show statistically significant improvement, but geographic coverage expands from 9 to 36 countries, and the cluster boundaries are defined by the same variables used in modelling.

E. *Feature Importance and Threshold Effects*

Globally, growing-season precipitation (permutation importance = 0.090) contributes more to predictions than heat stress days (0.056). However, the dominant stressor varies by cluster: precipitation drives yield variability in semi-arid (Cluster 0) and tropical (Cluster 2) systems, while heat stress days dominate in temperate Europe (Cluster 1).

Partial dependence plots (Fig. 4) reveal clear non-linear threshold effects. Maize yield predictions remain near average up to approximately 50 heat stress days per growing season, beyond which predicted yield drops sharply and stabilises at a lower level. For precipitation, yield suffers most when growing-season rainfall falls between 400–600 mm (moderate drought), and additional rainfall above approximately 1,000 mm provides diminishing returns.

TABLE II
PERMUTATION FEATURE IMPORTANCE (GLOBAL RF, 36 COUNTRIES)

Feature	Importance (mean)	Dominant in Cluster
Growing-season precipitation	0.090	0 (Semi-Arid), 2 (Tropical)
Heat stress days	0.056	1 (Temperate)

IV. DISCUSSION

The strong Kruskal-Wallis result ($H = 1195.22$) confirms that data-driven climate clustering captures real differences in maize yield instability, a result that theory-based panels would not have guaranteed without empirical verification. The finding that the tropical cluster exhibits the highest instability (mean

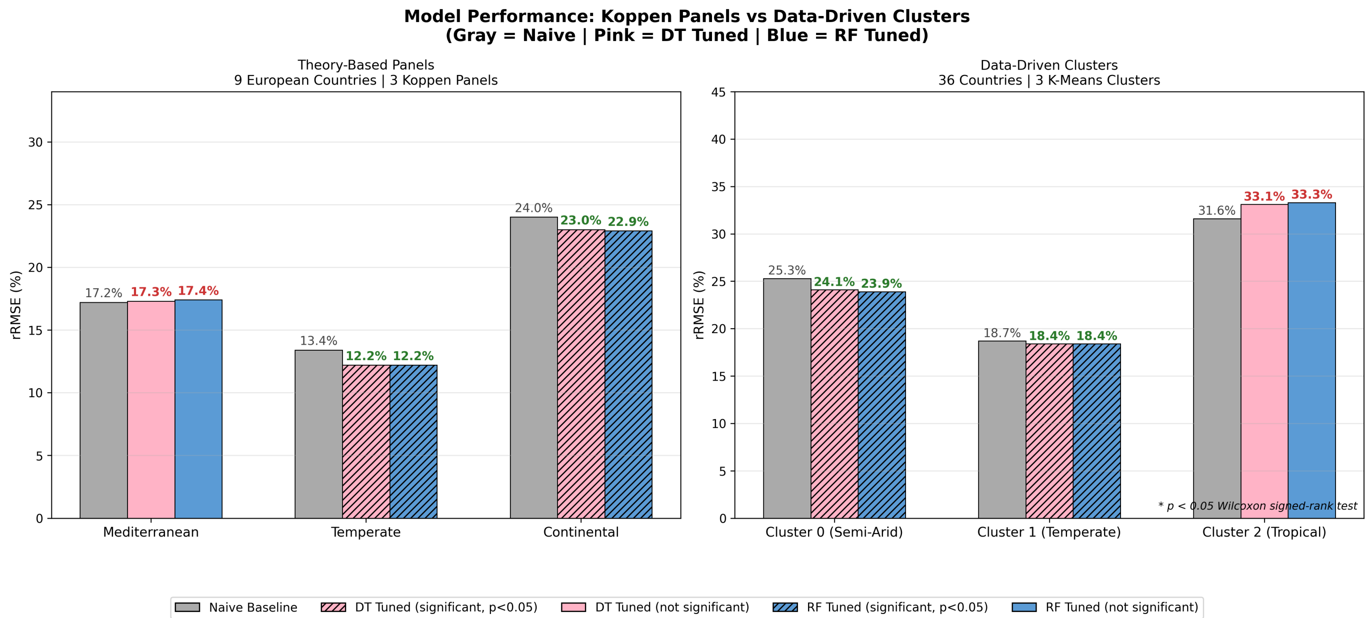

Fig. 3. Comparison of model performance (Naive, tuned DT, tuned RF) under theory-based Köppen panels (left, 9 European countries) vs data-driven k-means clusters (right, 36 countries). Hatching indicates statistically significant improvement over the naive baseline ($p < 0.05$, Wilcoxon signed-rank test).

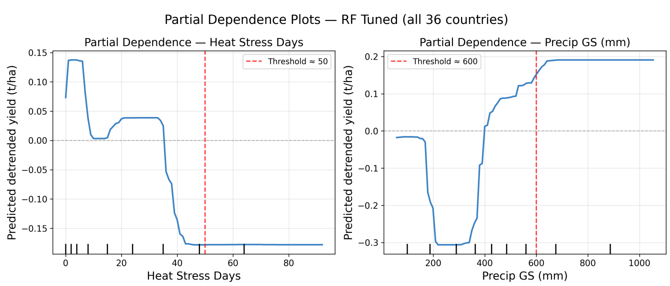

Fig. 4. Partial dependence plots for heat stress days (left) and growing-season precipitation (right) in original units. Dashed lines indicate key thresholds: ∼50 heat stress days and ∼600 mm precipitation.

CV = 27.5%) is consistent with literature on climate variability and food security in sub-Saharan Africa [2], where rainfall unpredictability is a primary driver of yield risk.

The modest predictive improvement over the naive baseline (1.3–0.7 percentage points rRMSE reduction in significant clusters) reflects an important finding: two climate variables explain only a limited share of inter-annual yield variance. This is consistent with Lobell and Field [1], who found that climate explains a smaller fraction of yield variability than yield levels suggest. Non-climatic factors; irrigation infrastructure, fertiliser use, cultivar selection, and soil quality act as buffers or amplifiers that the current two-feature model cannot capture.

The threshold effects revealed by partial dependence analysis are particularly meaningful. The sharp yield decline beyond 50 heat stress days aligns with known crop physiology: maize pollination and grain filling are acutely sensitive to temperatures above 30°C [4], and once a critical number of exposure days is exceeded, yield damage is largely irreversible within a growing season. The precipitation trough at 400–600 mm suggests that moderate drought, not extreme drought, produces the most widespread damage, possibly because irrigation partially offsets severe deficits in the most arid systems in the dataset.

The divergence in dominant stressors between clusters carries direct implications for adaptation. In temperate Europe (Cluster 1), managing episodic heat events through heat-tolerant cultivars and irrigation scheduling is the priority. In semi-arid and tropical systems (Clusters 0 and 2), water availability is the binding constraint. A single globally-pooled adaptation recommendation would be misleading for at least one of these contexts.

A. Limitations

Several limitations should be noted. The two-variable feature set excludes vapour pressure deficit, growing degree days, and soil moisture predictors known to improve crop model performance. Brazil and the United States were excluded from ML training due to sampling dominance, leaving Cluster 2 under-represented with only four training countries. The CY-Bench dataset does not distinguish rainfed from irrigated production, which confounds the interpretation of precipitation effects in heavily irrigated regions.

V. Conclusion

This study demonstrates that data-driven climate clustering outperforms theory-based regional panels for analysing maize yield instability at global scale. Hemisphere-aware feature engineering, non-parametric statistical testing, and partial dependence analysis together provide a reproducible, methodologically transparent framework applicable to other crops and datasets. The threshold-based climate-yield relationships identified here, 50 heat stress days and 600 mm precipitation, offer concrete, actionable benchmarks for climate adaptation planning.

AI Use Statement

Artificial intelligence was used to assist with code development and language check for drafting parts of the report. All analytical decisions, research questions, and interpretations were made by the author.

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